LABORATORY REPORT · ELECTROCHEMISTRY / THIN-FILM FABRICATION

Electrochemical Deposition of Copper on Graphite Electrodes

Item	Details
Project type	Side project within a device-fabrication project led by the group's postdoctoral researcher
Institution	Shahid Beheshti University – research group of Prof. Majid Mohseni
Role	Execution of the deposition workflow: electrolyte preparation, cell assembly, deposition run, post-processing and a basic electrical check of the coating
System	$CuSO_4$ / SDS aqueous electrolyte; two graphite electrodes; 10 V DC

1. Abstract

Matte copper coatings were deposited on graphite electrodes from a dilute $CuSO_4$ electrolyte containing sodium dodecyl sulfate (SDS) as a surfactant additive. A two-electrode cell was operated at 10 V DC (current limit 1.5 A) for more than 30 minutes, with copper plating onto the graphite cathode. After deposition the electrodes were rinsed with organic solvent and dried by heating. A matte, red-brown copper coating formed on the cathode, and two-probe multimeter measurements confirmed that it is electrically conductive. The matte finish is consistent with fast, transport-limited growth under a high applied bias and a low Cu^{2+} concentration.

2. Objective

The objective was to deposit a conductive copper coating on graphite electrodes by simple DC electrodeposition from a dilute, additive-containing electrolyte, and to verify the electrical conductivity of the resulting coating.

3. Materials and Equipment

- **Copper source:** copper(II) sulfate pentahydrate ($CuSO_4 \cdot 5H_2O$).
- **Additive:** sodium dodecyl sulfate (SDS), an anionic surfactant.
- **Electrodes:** two graphite plates (cathode and counter electrode).
- **Cell:** 200 mL beaker with a custom electrode holder on a retort stand.
- **Power source:** programmable DC power supply (30 V / 5 A).
- **Preparation tools:** analytical balance (0.1 mg resolution) and ultrasonic bath.
- **Post-processing:** organic solvents (ethanol/acetone) and a heat source for drying.
- **Characterisation:** multimeter (two-probe measurement).

4. Experimental Procedure

4.1 Electrolyte preparation

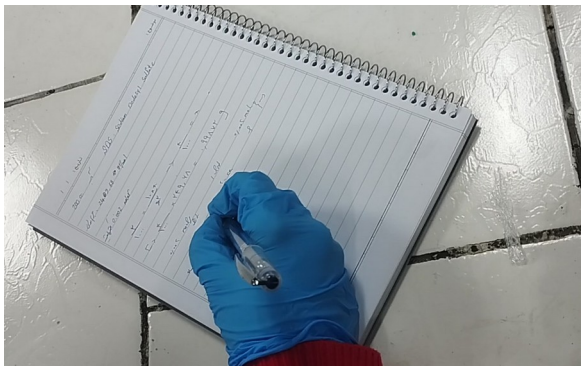
The electrolyte was a dilute (millimolar) aqueous solution of about 200 mL containing Cu^{2+} from copper(II) sulfate pentahydrate and SDS as a surfactant additive. Reagent masses were calculated from the target

concentrations (Table 1; Figure 1a), weighed on an analytical balance (Figure 1b) and dissolved in an ultrasonic bath (Figure 1c).

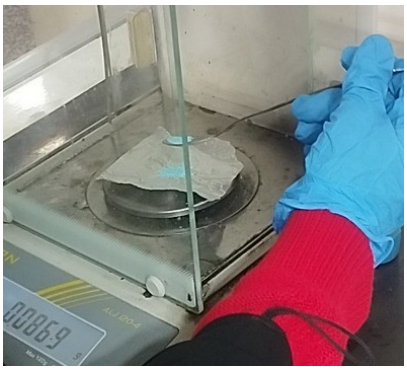
Component	Target concentration	Mass (≈)	Role
$CuSO_4 \cdot 5H_2O$	≈ 1.7 mM (200 mL)	0.087 g	Cu^{2+} source
SDS (M = 288.38 g/mol)	≈ 2 mM (200 mL)	0.115 g	Anionic surfactant additive

Table 1. Electrolyte composition (≈ 200 mL aqueous solution).

The surfactant improves electrode wetting and influences the morphology of the deposit.



(a) Notebook calculation of SDS concentration and mass



(b) Weighing $CuSO_4 \cdot 5H_2O$ on the analytical balance



(c) Dissolution in the ultrasonic bath

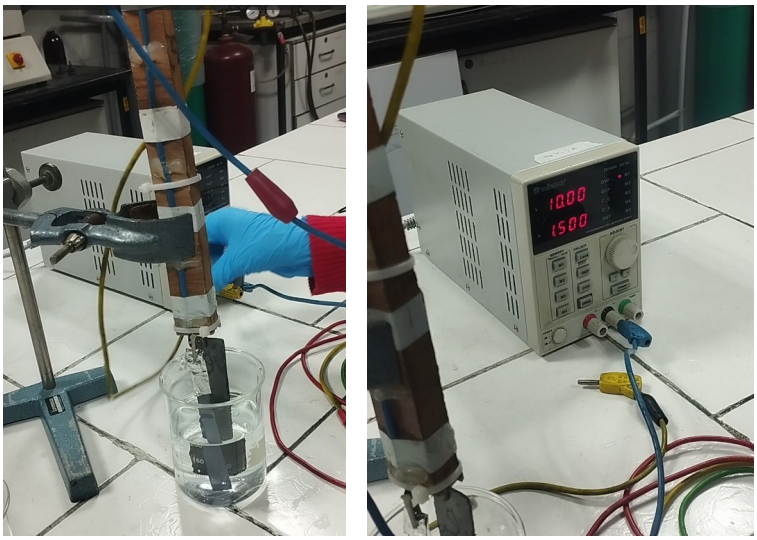
Figure 1. *Electrolyte preparation.*

4.2 Electrochemical cell and deposition

Deposition was carried out in a two-electrode cell at 10 V DC for more than 30 minutes, with copper plating onto the graphite cathode (Figure 2). The cell parameters are listed in Table 2.

Parameter	Value
Configuration	Two-electrode cell, 200 mL beaker
Cathode (substrate)	Graphite plate
Counter electrode	Graphite plate
Electrode mounting	Custom holder on a retort stand
Power source	Programmable DC power supply (30 V / 5 A)
Applied bias	10 V DC (current limit 1.5 A)
Duration	> 30 min

Table 2. Deposition cell and process parameters.



(a) Graphite electrodes immersed in the electrolyte (b) Power supply set to 10 V, 1.5 A limit

Figure 2. *Two-electrode deposition setup.*

4.3 Post-processing

After deposition, the coated electrodes were rinsed with organic solvent (ethanol/acetone) to remove residual electrolyte and surfactant, then dried by heating (Figure 3).

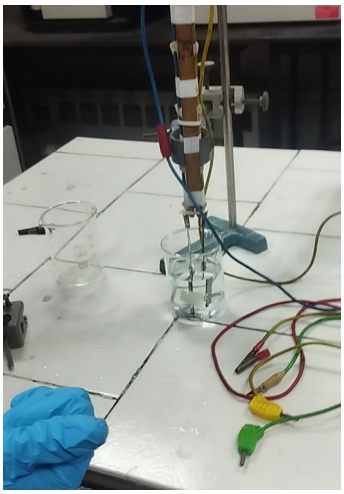


Figure 3. *Rinsing stage after deposition.*

5. Results and Discussion

A matte, red-brown copper coating formed on the graphite cathode, and two-probe multimeter measurements confirmed that it is electrically conductive.

The matte, red-brown finish (rather than a bright one) is consistent with fast, transport-limited growth. A high applied bias combined with a low Cu^{2+} concentration typically yields a rough, granular deposit. Quantitative data such as coating thickness, current–time behaviour and surface morphology were not recorded in this work.

6. Recommendations for Improvement

Area	Recommended improvement
Potential control	Use a three-electrode cell with a potentiostat so that the cathode potential is set directly rather than the total cell voltage.
Thickness estimate	Log current over time and estimate the deposited thickness from Faraday's law.
Substrate pretreatment	Polish and ultrasonically clean the graphite to improve adhesion and uniformity.
Characterisation	SEM and XRD for morphology and crystallinity; four-point probe for sheet resistance.
Electrolyte depletion	With an inert graphite anode, consumed Cu^{2+} is not replenished, unlike with a soluble copper anode; this should be considered for longer runs.

Table 3. Recommended improvements.

7. Conclusion

A conductive, matte copper coating was successfully deposited on graphite electrodes from a dilute $CuSO_4$/SDS electrolyte using a simple two-electrode DC cell. The morphology is consistent with fast, transport-limited growth. Potential-controlled deposition, thickness estimation and structural and electrical characterisation are the main steps needed to turn this qualitative result into a quantitative, reproducible process.

8. Skills Demonstrated

- Electrolyte design and stoichiometric calculations
- Electrochemical cell assembly and DC power-supply operation
- Substrate post-processing (solvent rinsing, thermal drying)
- Basic electrical characterisation of thin coatings
- Working within a multi-person fabrication project